EXPLICIT FEIGIN-ODESSKII BRACKETS ON PROJECTIVE FIVE-SPACE

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Allowed standard prerequisites (editorial).

- Research prerequisites named in the original article. Full original bibliography retained; all source-local definitions and supporting arguments retained.

1. Introduction

The goal of this paper is to present explicit formulas for certain algebraic Poisson brackets on $\mathbb{P}^5$.

Recall that two Poisson brackets $\{\cdot,\cdot\}_1$, $\{\cdot,\cdot\}_2$ are called *compatible* if any linear combination $\{\cdot,\cdot\}_1 + \lambda \cdot \{\cdot,\cdot\}_2$ is still a Poisson bracket (i.e., satisfies the Jacobi identity). Pairs of compatible Poisson brackets play an important role in the theory of integrable systems.

With every normal elliptic curve C in $\mathbb{P}^n$ one can associate naturally a Poisson bracket on $\mathbb{P}^n$, called a *Feigin–Odesskii bracket of type* $q_{n+1,1}$. The corresponding quadratic Poisson brackets on $\mathbb{A}^{n+1}$ arise as quasi-classical limits of Feigin–Odesskii elliptic algebras. On the other hand, they can be constructed using the geometry of vector bundles on C (see [2, 8]).

It was discovered by Odesskii–Wolf [6] that for every n there exists a family of 9 linearly independent mutually compatible Poisson brackets on $\mathbb{P}^n$, such that their generic linear combinations are Feigin–Odesskii brackets of type $q_{n+1,1}$. In [3], this construction was explained and extended in terms of anticanonical line bundles on del Pezzo surfaces. It was observed in [3, Example 4.6] that in this framework one also obtains 10 linearly independent mutually compatible Poisson brackets on $\mathbb{P}^5$. In this paper, we will produce explicit formulas for these 10 brackets (see Theorem 3.2).

2. Homological perturbation for $\mathbb{P}^n$

2.1. Formula for the homotopy.

Let

$$H = \bigoplus_{p\geq 0,\, q\in\mathbb{Z}} H^p(\mathbb{P}^n, \mathcal{O}(q))$$

be the cohomology algebra of line bundles on $\mathbb{P}^n$, and

$$A = \Big(\bigoplus_{p\geq 0,\, q\in\mathbb{Z}} C^p(\mathbb{P}^n, \mathcal{O}(q)), d \Big)$$

the Čech complex with respect to the standard open covering $U_i = (x_i \neq 0)$ of $\mathbb{P}^n$. There is a natural dg-algebra structure on A, such that the corresponding cohomology algebra is H. The multiplication on A is defined as follows. For $\alpha \in C^p(\mathbb{P}^n, \mathcal{O}(q))$ and $\beta \in C^{p'}(\mathbb{P}^n, \mathcal{O}(q'))$, we define $\alpha\beta \in C^{p+p'}(\mathbb{P}^n, \mathcal{O}(q+q'))$ by

$$(\alpha\beta)_{i_0 i_1 \ldots i_{p+p'}} := \alpha_{i_0 \ldots i_p}|_{U_{i_0 \ldots i_{p+p'}}} \cdot \beta_{i_p \ldots i_{p+p'}}|_{U_{i_0 \ldots i_{p+p'}}},$$

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where on the right hand side we use the multiplication map $\mathcal{O}(q)\otimes\mathcal{O}(q')\to\mathcal{O}(q+q')$.

The homological perturbation lemma equips H with a minimal A_∞-structure (m_n), where m_2 is the usual product on H. We will use the form of this lemma due to Kontsevich–Soibelman [5], which gives formulas for m_n as sums over trees. To apply homological perturbation, we need the following data:

- a projection $\pi\colon A\to H$,
- an inclusion $\iota\colon H\to A$, and
- a homotopy Q such that $\pi\iota=\mathrm{id}_H$ and $\mathrm{id}_A-\iota\pi=\mathrm{d}Q+Q\mathrm{d}$.

Recall that $H^0=\mathbb{C}[x_0,\dots,x_n]$,

$$H^n\simeq\bigoplus_{e_0,\dots,e_n<0}\mathbf{k}\cdot x_0^{e_0}x_1^{e_1}\cdots x_n^{e_n}\subset A^n,$$

and $H^i=0$ for $i\neq 0,n$. We define ι in degree zero by $\iota(f)_k=f$ for $k=0,1,\dots,n$. We define ι in degree n by $\iota(g)_{0\dots n}=g$. We define the projection in degree zero to be

$$\pi(\gamma)=\begin{cases}\gamma_n & \text{if } \gamma_n\in\mathbb{C}[x_0,\dots,x_n],\\ 0 & \text{else.}\end{cases}$$

To define π in degree n, we observe that

$$A^n=\bigoplus_{e_0,\dots,e_n\in\mathbb{Z}}\mathbf{k}\cdot x_0^{e_0}x_1^{e_1}\cdots x_n^{e_n},$$

and we let π be the natural projection to H^n.

To define the homotopy, we use that A decomposes as a direct sum of chain complexes

$$A=\oplus_{\vec{e}\in\mathbb{Z}^{n+1}}A(\vec{e}),$$

where $A(\vec{e})$ consists of all elements in A whose components are scalar multiples of $x^{\vec{e}}:=x_0^{e_0}x_1^{e_1}\cdots x_n^{e_n}$. In other words, $A(\vec{e})$ is the $\vec{e}$-isotypical summand with respect to the action of the group $\mathbb{G}_m^{n+1}$.

Let us set for $\vec{e}\in\mathbb{Z}^{n+1}$,

$$k(\vec{e}):=\max\{i\mid e_i\geq 0\}$$

(which is equal to $-\infty$ if all e_i are negative). There is then a standard homotopy Q defined on an element $\gamma\in A(\vec{e})^p$ by $Q(\gamma)_{i_0i_1\dots i_{p-1}}=\gamma_{k(\vec{e})i_0\dots i_{p-1}}$ if $k(\vec{e})>-\infty$ and $Q(\gamma)_{i_0i_1\dots i_{p-1}}=0$ otherwise (i.e., if all e_i are negative).

For a Laurent monomial $x^{\vec{e}}$ and a subset $I=\{i_0,\dots,i_p\}\subset\{0,1,\dots,n\}$ such that $I\supset\{0\leq i\leq n|e_i<0\}$, let us denote by $x_I^{\vec{e}}$ the element of A^p given by

$$(x_I^{\vec{e}})_{j_0\dots j_p}=\begin{cases}x^{\vec{e}} & \text{if } \{j_0,\dots,j_p\}=I,\\ 0 & \text{otherwise.}\end{cases}$$

Note that the condition $I\supset\{0\leq i\leq n\,|\,e_i<0\}$ guarantees that $x^{\vec{e}}$ is a regular section of the appropriate line bundle over $U_{i_0\dots i_p}$. Clearly, these elements form a basis for A and our homotopy operator Q is given by

$$Q\big(x_I^{\vec{e}}\big)=\begin{cases}(-1)^j x_{I\setminus k(\vec{e})}^{\vec{e}} & \text{if } k(\vec{e})=i_j\in I,\\ 0 & \text{otherwise.}\end{cases}$$

With these data one can in principle calculate all the higher products on the cohomology algebra H. Below, we will get explicit formulas in the case we need.

2.2. **Calculation of m_4 for $\mathbb{P}^2$.** We now specialize to the case of the projective plane $\mathbb{P}^2$. We have no higher products of odd degree because H and $H^{\otimes n}$ only live in even degrees. Also, for degree reasons the product m_4 will only be non-zero on elements $e \otimes f \otimes g \otimes h \in H^{\otimes 4}$ where one or two of the arguments lie in H^2 and the rest in H^0. Below, we will explicitly compute the product m_4 involving one argument in H^2. Thus, the following special case of the multiplication in A will be relevant: for a monomial $x^{\vec{e}}$ and a Laurent monomial $x^{\vec{e}'}$, we have

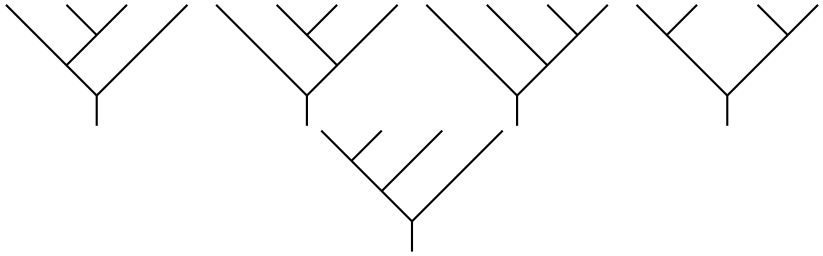
$$\iota_0\big(x^{\vec{e}}\big) \cdot x_I^{\vec{e}'} = x_I^{\vec{e}'} \cdot \iota_0\big(x^{\vec{e}}\big) = x_I^{\vec{e}+\vec{e}'}.$$

We use the formula

$$m_4(e,f,g,h) = -\sum_T \epsilon(T) m_T(e,f,g,h),$$

where the sum runs over all rooted binary trees with 4 leaves labeled e, f, g and h (from left to right). For each such tree T the expression $m_T(e,f,g,h)$ is computed by moving the inputs through that tree, applying ι at the leaves, applying the homotopy Q on each interior edge, multiplying elements of A at each inner vertex and finally applying the projection π at the bottom.

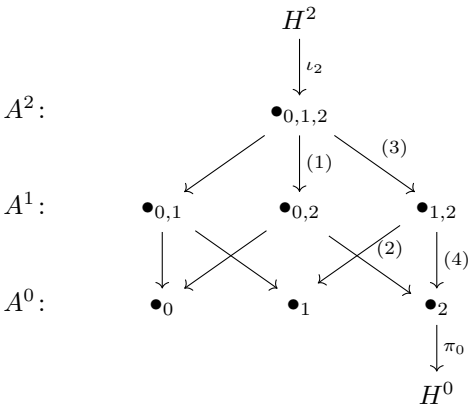
We have to sum over the following five trees, which we denote $T_1, \dots, T_5$, respectively,



Let us first consider the case $e \in H^2$ and $f, g, h \in H^0$ and let's take them all to be basis elements of H^2 and H^0:

$$e = \big(x_0^{\alpha_0} x_1^{\alpha_1} x_2^{\alpha_2}\big)_{\{0,1,2\}}, \qquad f = x_0^{a_0} x_1^{a_1} x_2^{a_2}, \qquad g = x_0^{b_0} x_1^{b_1} x_2^{b_2}, \qquad h = x_0^{c_0} x_1^{c_1} x_2^{c_2},$$

where $\alpha_0, \alpha_1, \alpha_2 < 0$ and $a_i, b_i, c_i \geq 0$ for $i = 0, 1, 2$. In this case only one of the trees above can be non-zero in the expression for $m_4(e,f,g,h)$, namely T_5, because in all other trees at some point the homotopy Q will be applied to an element of A^0. Below is a picture of the different summands in $A^\bullet$ and the possible ways the homotopy Q can map a monomial element in each summand:



When computing $m_{T_5}(e,f,g,h)$ we should move e through this diagram; at every node it gets multiplied by one of the other arguments and then it moves downwards

along one of the arrows. We see that we get non-zero result if we move either along (1) followed by (2) or along (3) followed by (4) (so that we land in $\bullet_2$). We claim that only the second route is possible. The reason is that at each node we multiply e by a monomial, so the exponents of x_0, x_1, x_2 will not decrease at any time. By the definition of Q, if e gets moved along (1) then after the multiplication at $\bullet_{0,1,2}$ the exponent of x_1 is non-negative while the exponent of x_2 is negative. Hence, after performing the multiplication at $\bullet_{0,2}$ the exponent of x_1 is still non-negative. It follows then from the definition of Q that e cannot move along (2) after moving along (1).

Now comes the computation of $m_{T_5}(e,f,g,h)$. Below, we denote by μ the multiplication in A. Then

$$\begin{aligned}
m_{T_5}(e,f,g,h) &= \pi\mu(Q\mu(Q\mu(e,f),g),h)\\
&= \pi\mu\big(Q\mu\big(Q\big(x_0^{\alpha_0+a_0}x_1^{\alpha_1+a_1}x_2^{\alpha_2+a_2}\big)_{\{0,1,2\}},g\big),h\big)\\
&\overset{(*)}{=} \pi\mu\big(Q\mu\big(\big(x_0^{\alpha_0+a_0}x_1^{\alpha_1+a_1}x_2^{\alpha_2+a_2}\big)_{\{1,2\}},g\big),h\big)\\
&= \pi\mu\big(Q\big(x_0^{\alpha_0+a_0+b_0}x_1^{\alpha_1+a_1+b_1}x_2^{\alpha_2+a_2+b_2}\big)_{\{1,2\}},h\big)\\
&\overset{(**)}{=} \pi\big(\mu\big(\big(x_0^{\alpha_0+a_0+b_0}x_1^{\alpha_1+a_1+b_1}x_2^{\alpha_2+a_2+b_2}\big)_{\{2\}},h\big)\big)\\
&= \pi\big(\big(x_0^{\alpha_0+a_0+b_0+c_0}x_1^{\alpha_1+a_1+b_1+c_1}x_2^{\alpha_2+a_2+b_2+c_2}\big)_{\{2\}}\big)\\
&\overset{(***)}{=} x_0^{\alpha_0+a_0+b_0+c_0}x_1^{\alpha_1+a_1+b_1+c_1}x_2^{\alpha_2+a_2+b_2+c_2},
\end{aligned}$$

where the symbols $(*)$, $(**)$ are $(***)$ mean that we get zero unless the following conditions hold:

$$(*)\ \begin{cases}\alpha_0+a_0\ge 0,\\ \alpha_1+a_1<0,\\ \alpha_2+a_2<0,\end{cases}\qquad (**)\ \begin{cases}\alpha_1+a_1+b_1\ge 0,\\ \alpha_2+a_2+b_2<0,\end{cases}\qquad (***)\ \begin{cases}\alpha_0+a_0+b_0+c_0\ge 0,\\ \alpha_1+a_1+b_1+c_1\ge 0,\\ \alpha_2+a_2+b_2+c_2\ge 0.\end{cases}$$

In the end, we have

$$m_4(e,f,g,h) = -m_{T_5}(e,f,g,h) = -\rho\big(\vec{\alpha};\vec{a},\vec{b},\vec{c}\big)\cdot x^{\vec{\alpha}+\vec{a}+\vec{b}+\vec{c}},$$

where

$$\rho\big(\vec{\alpha};\vec{a},\vec{b},\vec{c}\big) := \begin{cases}1 & \text{if } \alpha_0+a_0\ge 0,\ \alpha_1+a_1<0,\ \alpha_1+a_1+b_1\ge 0,\\ & \quad \alpha_2+a_2+b_2<0,\ \alpha_2+a_2+b_2+c_2\ge 0,\\ 0 & \text{else.}\end{cases}$$

Similarly, we compute m_4 applied to e, f, g, h in any given order. We have

$$\begin{aligned}
m_4(e,f,g,h) &= -\rho\big(\vec{\alpha};\vec{a},\vec{b},\vec{c}\big)\cdot x^{\vec{\alpha}+\vec{a}+\vec{b}+\vec{c}},\\
m_4(f,e,g,h) &= \big[-\rho\big(\vec{\alpha};\vec{a},\vec{b},\vec{c}\big)+\rho\big(\vec{\alpha};\vec{b},\vec{a},\vec{c}\big)-\rho\big(\vec{\alpha};\vec{b},\vec{c},\vec{a}\big)\big]\cdot x^{\vec{\alpha}+\vec{a}+\vec{b}+\vec{c}},\\
m_4(f,g,e,h) &= \big[\rho\big(\vec{\alpha};\vec{b},\vec{a},\vec{c}\big)-\rho\big(\vec{\alpha};\vec{b},\vec{c},\vec{a}\big)+\rho\big(\vec{\alpha};\vec{c},\vec{b},\vec{a}\big)\big]\cdot x^{\vec{\alpha}+\vec{a}+\vec{b}+\vec{c}},\\
m_4(f,g,h,e) &= \rho\big(\vec{\alpha};\vec{c},\vec{b},\vec{a}\big)\cdot x^{\vec{\alpha}+\vec{a}+\vec{b}+\vec{c}}.
\end{aligned}$$

3. Feigin–Odesskii brackets

3.1. **Bivectors on projective spaces.** It is well known that every $\mathbb{G}_m$-invariant bivector on a vector space V leads to a bivector on the projective space $\mathbb{P}V$. A bivector on V can be thought of as a skew-symmetric bracket $\{\cdot,\cdot\}$ on the polynomial algebra $S(V^*)$, which is a biderivation. Such a bracket is $\mathbb{G}_m$-invariant if and

only if the bracket of two linear forms is a quadratic form. In other words, such a bracket can be viewed as a skew-symmetric pairing

$$b\colon\ V^*\times V^*\to S^2(V^*).$$

The corresponding bivector Π on the projective space $\mathbb{P}V$ is determined by the skew-symmetric forms Π_v on $T^*_v\mathbb{P}V$ for each point $\langle v\rangle\in\mathbb{P}V$. We have an identification

$$T^*_v\mathbb{P}V=\langle v\rangle^\vee\subset V^*.$$

It is easy to see that under this identification we have

$$\Pi_v(s_1\wedge s_2)=b(s_1,s_2)(v), \tag{3.1}$$

where $s_1,s_2\in\langle v\rangle^\vee$. Here we take the value of the quadratic form $b(s_1\wedge s_2)$ at v.

We can use the above formula in reverse. Namely, suppose for some bivector Π on $\mathbb{P}V$ we found a skew-symmetric pairing b such that (3.1) holds. Then the $\mathbb{G}_m$-invariant bracket $\{\cdot,\cdot\}$ on $S(V)$ given by b induces the bivector Π on $\mathbb{P}V$. Note that if Π is a Poisson bivector on $\mathbb{P}V$, it is not guaranteed that the $\mathbb{G}_m$-invariant bracket $\{\cdot,\cdot\}$ on $S(V)$ is also Poisson, i.e., satisfies the Jacobi identity (but it is known that $\{\cdot,\cdot\}$ can be chosen to be Poisson, see [1, 7]).

3.2. **Recollections from** [3]**.** Below, we will denote simply by L_1L_2 the tensor product of line bundles L_1 and L_2.

Let ξ be a line bundle of degree n on an elliptic curve C. We fix a trivialization $\omega_C\simeq\mathcal{O}_C$. Then the associated Feigin–Odesskii Poisson structure Π (to which we will refer as *FO bracket*) on $\mathbb{P}H^1\big(\xi^{-1}\big)\simeq\mathbb{P}H^0(\xi)^*$ is given by the formula (see [3, Lemma 2.1])

$$\Pi_\phi(s_1\wedge s_2)=\langle\phi,\mathrm{MP}(s_1,\phi,s_2)\rangle, \tag{3.2}$$

where $\langle\phi\rangle\in\mathbb{P}\operatorname{Ext}^1(\xi,\mathcal{O})$, and $s_1,s_2\in\langle\phi\rangle^\perp$. Here we use the Serre duality pairing $\langle\cdot,\cdot\rangle$ between $H^0(\xi)$ and $H^1\big(\xi^{-1}\big)$ and the triple Massey product

$$\mathrm{MP}\colon\ H^0(\xi)\otimes H^1\big(\xi^{-1}\big)\otimes H^0(\xi)\to H^0(\xi)$$

that also agrees with the triple product m_3 obtained by homological perturbation from the natural dg enhancement of the derived category of coherent sheaves on C. There is some ambiguity in a choice of m_3 but for $s_1,s_2\in\langle\phi\rangle^\perp$, the expression in the right-hand side of (3.2) is well defined.

Next, assume that S is a smooth projective surface, L is a line bundle on S such that $H^*(S,LK_S)=0$, and let $C\subset S$ be a smooth connected anticanonical divisor (which is an elliptic curve), so we have an exact sequence of coherent sheaves on S,

$$0\to K_S\xrightarrow{F}\mathcal{O}_S\to\mathcal{O}_C\to 0. \tag{3.3}$$

We have a natural restriction map

$$H^0(S,L)\to H^0(C,L|_C).$$

The exact sequence

$$0\to LK_S\xrightarrow{F}L\to L_C\to 0 \tag{3.4}$$

shows that under our assumptions this restriction map is an isomorphism.

Thus, the FO bracket on $\mathbb{P}H^0(L|_C)^*$ associated with $(C,L|_C)$ (defined up to rescaling) can be viewed as a Poisson structure on a fixed projective space $\mathbb{P}V^*$, where

$$V:=H^0(S,L).$$

By [3, Theorem 4.4], the Poisson brackets on $\mathbb{P}V^*$ associated with different anticanonical divisors are compatible. More precisely, we get a linear map from

$H^0\big(S, K_S^{-1}\big)$ to the space of bivectors on $\mathbb{P}V^*$, whose image lies in the space of Poisson brackets.

3.3. **Feigin–Odesskii bracket for an anticanonical divisor.** We keep the data (S, L) of the previous subsection. Let $i\colon C \hookrightarrow S$ be an anticanonical divisor in S, with the equation $F \in H^0\big(S, K_S^{-1}\big)$. We want to write a formula for the FO bracket $\Pi = \Pi_F$ on $\mathbb{P}V^*$ in terms of higher products on the surface S and the equation F. For this we rewrite the right-hand side of formula (3.2). Let us write the triple product in this formula as MP^C to remember that it is defined for the derived category of C.

Proposition 3.1.

(*i*) *In the above situation, given $e \in V^*$ and $s_1, s_2 \in \langle e\rangle^\perp$, one has*

$$\big\langle e, \mathrm{MP}^C(s_1|_C, e, s_2|_C)\big\rangle = \langle m_4(F, s_1, e, s_2) - m_4(s_1, F, e, s_2), e\rangle,$$

where we use the identification $V^* \simeq H^2\big(S, L^{-1}K_S\big)$ given by Serre duality and consider the A_∞-products on S,

$$m_4\colon\ H^0\big(K_S^{-1}\big)H^0(L)H^2\big(L^{-1}K_S\big)H^0(L) \to H^0(L),$$
$$H^0(L)H^0\big(K_S^{-1}\big)H^2\big(L^{-1}\big)H^0(L) \to H^0(L),$$

obtained by the homological perturbation.

(*ii*) *Assume that a generic anticanonical divisor is smooth (and connected). Then*

$$\Pi_F|_e(s_1 \wedge s_2) := \langle m_4(F, s_1, e, s_2) - m_4(s_1, F, e, s_2), e\rangle$$

gives a collection of compatible Poisson brackets on $\mathbb{P}V$ depending linearly on F.

Proof. (i) By Serre duality, $H^*\big(S, L^{-1}\big) = 0$, so the map

$$H^1\big(C, L^{-1}\big|_C\big) \to H^2\big(S, L^{-1}K_S\big),$$

induced by the exact sequence

$$0 \to L^{-1}K_S \to L^{-1} \to L^{-1}\big|_C \to 0,$$

is an isomorphism. It is a standard fact that this isomorphism is the dual to the isomorphism $H^0(S, L) \to H^0(C, L|_C)$ given by the restriction, via Serre dualities on S and C. Let us denote by $e_C \in H^1\big(C, L^{-1}\big|_C\big)$ the element corresponding to $e \in H^2\big(S, L^{-1}K_S\big)$ under the above isomorphism.

We claim that the triple Massey product $\mathrm{MP}^C(s_1|_C, e_C, s_2|_C) = m_3(s_1|_C, e_C, s_2|_C)$ corresponding to the arrows

$$\mathcal{O}_C \xrightarrow{s_2|_C} L|_C \xrightarrow{e_C} \mathcal{O}_C \xrightarrow{s_1|_C} L|_C$$

(where the middle arrow has degree 1) agrees with the corresponding triple Massey product on S,

$$\mathcal{O}_S \xrightarrow{s_2} L \xrightarrow{e_C} \mathcal{O}_C \xrightarrow{s_1|_C} L|_C.$$

Indeed, the relevant spaces are identified via the restriction maps. Let

$$r\colon\ \mathcal{O}_S \to \mathcal{O}_C, \qquad r_L\colon\ L \to L|_C$$

be the natural maps. Then we have to check that for $s_1, s_2 \in \langle e\rangle^\perp \subset H^0(S, L)$, one has

$$m_3(s_1|_C, e_C, s_2|_C)r \equiv m_3(s_1|_C, e_C r_L, s_2) \mod \langle s_1|_C r, s_2|_C r\rangle,$$

where we view this as equality of cosets in $\mathrm{Hom}(\mathcal{O}_S, L|_C)$. The A_∞-identities imply that

$$m_3(s_1|_C, e_C, s_2|_C)r = m_3(s_1|_C, e_C, s_2|_C r) \pm s_1|_C m_3(e_C, s_2|_C, r),$$

where $s_2|_C r = r_L s_2$, and

$$m_3(s_1|_C, e_C, r_L s_2) = m_3(s_1|_C, e_C r_L, s_2) \pm s_1|_C m_3(e_C, r_L, s_2) \pm m_3(s_1|_C, e_C, r_L)s_2.$$

Combining these two identities, we deduce our claim.

Thus, it is enough to calculate the Massey product $\mathrm{MP}(s_1|_C, e_C r_L, s_2)$. Using the exact sequences (3.3) and (3.4), we can represent $\mathcal{O}_C$ (resp. L_C) by the twisted complex $[K_S[1] \to \mathcal{O}_S]$ (resp. $[LK_S[1] \to L]$).

In terms of these resolutions, the elements of $\mathrm{Ext}^1(L, \mathcal{O}_C)$ get represented by $\mathrm{Ext}^2(L, K_S) \subset \mathrm{hom}^\bullet(L, [K_S[1] \to \mathcal{O}_S])$, while the element of $\mathrm{Hom}(\mathcal{O}_C, L|_C)$ corresponding to $s \in H^0(S, L) \simeq H^0(C, L|_C)$ is given by the natural map of twisted complexes induced by the multiplication by s. The elements of $\mathrm{Hom}(\mathcal{O}_S, L|_C)$ are identified with $\mathrm{Hom}(\mathcal{O}_S, L) \simeq \mathrm{hom}^0(\mathcal{O}_S, [LK_S[1] \to L])$. Thus, the m_3 product we are interested is given by the following triple product in the category of twisted complexes over S:

$$\begin{array}{ccc}
\mathcal{O}_S & & \\
\downarrow{\scriptstyle s_2} & & \\
L & & \\
\downarrow{\scriptstyle e} & & \\
K_S[1] & \xrightarrow{F} & \mathcal{O}_S \\
\downarrow{\scriptstyle s_1} & & \downarrow{\scriptstyle s_1} \\
LK_S[1] & \xrightarrow{F} & L,
\end{array}$$

where we view e as a morphism of degree 1 from L to $K_S[1]$. Now the formula for m_3 on twisted complexes (see [4, Section 7.6]) gives

$$m_4(F, s_1, e, s_2) - m_4(s_1, F, e, s_2)$$

(here the insertions of F correspond to insertions of the differentials in the twisted complexes).

(ii) It is clear that Π_F gives a linear map from $H^0\big(S, \omega_S^{-1}\big)$ to the space of bivectors on $\mathbb{P}V$. By (i), for generic F we get a Poisson bracket. Hence, this is true for all F. □

3.4. **The case leading to 10 compatible brackets on $\mathbb{P}^5$.** We can apply Proposition 3.1 to the case $S = \mathbb{P}^2$ and $L = \mathcal{O}(2)$. Note that the assumptions are satisfied in this case since $LK_S = \mathcal{O}(-1)$ has vanishing cohomology. Thus, for each $F \in H^0\big(\mathbb{P}^2, \mathcal{O}(3)\big)$ giving a smooth cubic, we get a formula for the FO-bracket Π_F on $\mathbb{P}H^0\big(\mathbb{P}^2, \mathcal{O}(2)\big)^* = \mathbb{P}^5$. Hence, we get a family of 10 (the dimension of $H^0\big(\mathbb{P}^2, \mathcal{O}(3)\big)$ compatible brackets on $\mathbb{P}^5$ (we also know this from [3, Proposition 4.7]). The fact that these 10 brackets are linearly independent follows from the compatibility of this construction with the GL_3-action and is explained in [3, Proposition 4.7].

Now we will derive formulas for the brackets $\{\,,\,\}_F$ on the algebra of polynomials in 6 variables which induce the above Poisson brackets on $\mathbb{P}V \simeq \mathbb{P}^5$, where

$$V = H^0\big(\mathbb{P}^2, \mathcal{O}(2)\big)^*.$$

They depend linearly on F, so we will just give formulas for $\{\ ,\ \}_{x^{\vec{c}}}$, where $x^{\vec{c}}$ runs through all 10 monomials of degree 3 in (x_0, x_1, x_2).

Let us set

$$\Delta(n) := \begin{cases} \{(a_0,a_1,a_2) \in \mathbb{Z}^3 \mid a_0+a_1+a_2 = n,\ a_i \geq 0 \text{ for } i=0,1,2\} & \text{if } n \geq 0, \\ \{(\alpha_0,\alpha_1,\alpha_2) \in \mathbb{Z}^3 \mid \alpha_0+\alpha_1+\alpha_2 = n,\ \alpha_i < 0 \text{ for } i=0,1,2\} & \text{if } n < 0. \end{cases}$$

Note that $\{x^{\vec{e}} \mid \vec{e} \in \Delta(n)\}$ forms a basis for $H^0\big(\mathbb{P}^2, \mathcal{O}(n)\big)$ when $n \geq 0$, while $\big\{x^{\vec{e}}_{\{0,1,2\}} \mid \vec{e} \in \Delta(n)\big\}$ is a basis for $H^2\big(\mathbb{P}^2, \mathcal{O}(n)\big)$ when $n < 0$. In particular, we use $\big\{x^{\vec{a}} \mid \vec{a} \in \Delta(2)\big\}$ as a basis in $V^* = H^0\big(\mathbb{P}^2, \mathcal{O}(2)\big)$. Our brackets should associate to a pair of elements of this basis a quadratic form in the same variables.

Theorem 3.2. *One has for $\vec{a}, \vec{b} \in \Delta(2)$, $\vec{c} \in \Delta(3)$,*

$$\big\{x^{\vec{a}}, x^{\vec{b}}\big\}_{x^{\vec{c}}} := \sum_{\vec{a}\,', \vec{b}\,' \in \Delta(2)} \Big[\sum_{\sigma} -\operatorname{sgn}(\sigma) \tilde{\rho}\big(\sigma\vec{a}, \sigma\vec{b}, \sigma\vec{c}, \vec{a}\,', \vec{b}\,'\big)\Big] x^{\vec{a}\,'} x^{\vec{b}\,'}, \tag{3.5}$$

where the second sum is over the symmetric group on the letters $\{a, b, c\}$ and

$$\tilde{\rho}\big(\vec{a}, \vec{b}, \vec{c}, \vec{a}\,', \vec{b}\,'\big) := \begin{cases} 1 & \text{if } a_0' \leq a_0 - 1,\ a_1' > a_1 - 1,\ a_1' \leq a_1 + b_1 - 1, \\ & \quad a_2 + b_2 < a_2' + 1,\ c_2 + a_2 + b_2 \geq a_2' + 1, \\ & \quad a_0' + b_0' = a_0 + b_0 + c_0 - 1,\ a_1' + b_1' = a_1 + b_1 + c_1 - 1, \\ 0 & \text{else.} \end{cases}$$

Proof. By Serre duality, we can identify $V = H^0\big(\mathbb{P}^2, \mathcal{O}(2)\big)^*$ with $H^2\big(\mathbb{P}^2, \mathcal{O}(-5)\big)$. By Proposition 3.1, the bracket $\{x^{\vec{a}}, x^{\vec{b}}\}_{x^{\vec{c}}}$ is the quadratic form on $V \simeq H^2\big(\mathbb{P}^2, \mathcal{O}(-5)\big)$ given by

$$Q(e) := \big\langle e, m_4\big(x^{\vec{c}}, x^{\vec{a}}, e, x^{\vec{b}}\big) - m_4\big(x^{\vec{a}}, x^{\vec{c}}, e, x^{\vec{b}}\big)\big\rangle.$$

We can write

$$e = \sum_{\vec{\alpha} \in \Delta(-5)} c_{\vec{\alpha}} x^{\vec{\alpha}}_{\{0,1,2\}} \in H^2\big(\mathbb{P}^2, \mathcal{O}(-5)\big).$$

Using the formulas for m_4 from the end of Section 2.2, we get

$$Q(e) = \sum_{\vec{\alpha}, \vec{\beta} \in \Delta(-5)} \Big[\sum_{\sigma} -\operatorname{sgn}(\sigma) \rho\big(\vec{\alpha}; \sigma\vec{a}, \sigma\vec{b}, \sigma\vec{c}\big)\Big] \delta\big(\vec{\alpha}, \vec{\beta}, \vec{a}, \vec{b}, \vec{c}\big) c_{\vec{\alpha}} c_{\vec{\beta}},$$

where the second sum runs over the symmetric group on the letters $\{a, b, c\}$ and

$$\delta(\vec{\alpha}, \vec{\beta}, \vec{a}, \vec{b}, \vec{c}) = \begin{cases} 1 & \text{if } \vec{\alpha} + \vec{\beta} + \vec{a} + \vec{b} + \vec{c} = (-1,-1,-1), \\ 0 & \text{else.} \end{cases}$$

We have to show that the element in $S^2\big(H^0\big(\mathbb{P}^2, \mathcal{O}(2)\big)\big)$ given by the right-hand side of (3.5) defines the same quadratic form Q on $H^2\big(\mathbb{P}^2, \mathcal{O}(-5)\big)$. To see this, we apply it to the element $e = \sum_{\vec{\alpha} \in \Delta(-5)} c_{\vec{\alpha}} x^{\vec{\alpha}}_{\{0,1,2\}} \in H^2\big(\mathbb{P}^2, \mathcal{O}(-5)\big)$. For $\vec{\alpha} \in \Delta(-5)$, we set $\vec{\alpha}^* := (-1,-1,-1) - \vec{\alpha}$ and then we compute

$$\begin{aligned} &\Big(\sum_{\vec{a}\,', \vec{b}\,' \in \Delta(2)} \Big[\sum_{\sigma} -\operatorname{sgn}(\sigma) \tilde{\rho}\big(\sigma\vec{a}, \sigma\vec{b}, \sigma\vec{c}, \vec{a}\,', \vec{b}\,'\big)\Big] x^{\vec{a}\,'} x^{\vec{b}\,'}\Big)(e) \\ = &\sum_{\vec{\alpha}, \vec{\beta} \in \Delta(-5)} \sum_{\vec{a}\,', \vec{b}\,' \in \Delta(2)} \Big[\sum_{\sigma} -\operatorname{sgn}(\sigma) \tilde{\rho}\big(\sigma\vec{a}, \sigma\vec{b}, \sigma\vec{c}, \vec{a}\,', \vec{b}\,'\big)\Big] \langle x^{\vec{a}\,'}, x^{\vec{\alpha}}_{\{0,1,2\}}\rangle \langle x^{\vec{b}\,'}, x^{\vec{\beta}}_{\{0,1,2\}}\rangle c_{\vec{\alpha}} c_{\vec{\beta}} \\ = &\sum_{\vec{\alpha}, \vec{\beta} \in \Delta(-5)} \Big[\sum_{\sigma} -\operatorname{sgn}(\sigma) \tilde{\rho}\big(\sigma\vec{a}, \sigma\vec{b}, \sigma\vec{c}, \vec{\alpha}^*, \vec{\beta}^*\big)\Big] c_{\vec{\alpha}} c_{\vec{\beta}}. \end{aligned}$$

Now it only remains to note that for any permutation σ, one has

$$\tilde{\rho}\big(\sigma\vec{a},\sigma\vec{b},\sigma\vec{c},\vec{\alpha}^*,\vec{\beta}^*\big)=\rho\big(\vec{\alpha};\sigma\vec{a},\sigma\vec{b},\sigma\vec{c}\big)\delta\big(\vec{\alpha},\vec{\beta},\vec{a},\vec{b},\vec{c}\big),$$

for $\tilde{\rho}$ given in the formulation of the theorem. □

Remarks 3.3.

(1) Note that when we take $\vec{c}=(0,0,3)$ only two permutations σ, namely, $\sigma=1$ and $\sigma=(a\ b)$, can give non-zero terms in the formula of Theorem 3.2. When $\vec{c}=(1,2,0)$ all permutations except $\sigma=1$ and $\sigma=(a\ b)$ may give non-zero terms. When $\vec{c}=(1,1,1)$ all permutations can give non-zero terms.

(2) It is not true that formulas (3.5) define compatible Poisson brackets on the algebra of polynomials in 6 variables: this is true only for the induced brackets on $\mathbb{P}^5$ (in other words, the relevant identities hold only for the ratios of coordinates x_i/x_j).

Acknowledgements. We thank the anonymous referee for useful remarks. A.P. is partially supported by the NSF grant DMS-2001224, and within the framework of the HSE University Basic Research Program and by the Russian Academic Excellence Project '5-100'.

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